

Lesson Objective

Interpret the unknown in multiplication and division to model and solve problems using units of 6 and 7.

Materials

(S) Personal white board

PROBLEM

- 1 Model a multiplication word problem involving units of 6 by drawing and labeling a tape diagram, then solve by breaking the factor into two smaller facts.**

Rosa has 7 boxes of crayons. Each box has 6 crayons. How many crayons does Rosa have in all?

Draw and label a tape diagram to model the problem. Write an equation using a letter for the unknown. Solve.

Model a multiplication word problem involving units of 6 by drawing and labeling a tape diagram, then solve by breaking the factor into two smaller facts.

Rosa has 7 boxes of crayons. Each box has 6 crayons. How many crayons does Rosa have in all?

Draw and label a tape diagram to model the problem. Write an equation using a letter for the unknown. Solve.

Answer: $c = 42$ crayons; $7 \times 6 = c$; $(5 \times 6) + (2 \times 6) = 30 + 12 = 42$

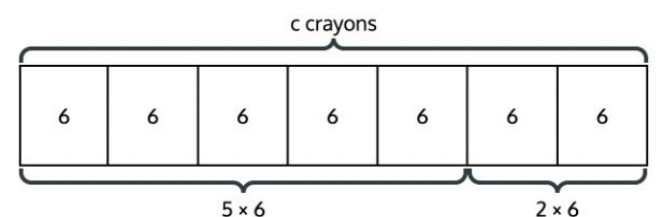
TEACHER GUIDANCE

Tell students that we can draw a tape diagram to model this problem. Guide students through drawing a tape diagram with 7 equal parts, each labeled "6 crayons," and a bracket on top labeled "c crayons."

Ask students what the letter c represents in this problem. (The total number of crayons.)

Guide students to write the equation $7 \times 6 = c$. Ask students if they know a way to break 7×6 into two smaller facts they already know. Model partitioning the tape diagram into a group of 5 and a group of 2, and write $(5 \times 6) + (2 \times 6)$.

Work with students to solve: $30 + 12 = 42$. Have students write $c = 42$ crayons and a sentence to answer the problem.



PROBLEM

- 2 **Model a division word problem involving units of 6 by drawing a tape diagram with an unknown number of groups, then solve using a related multiplication strategy or by decomposing the dividend.**

Marcus has 54 stickers. He puts 6 stickers on each page. How many pages does Marcus use?

Draw and label a tape diagram to model the problem. Write an equation using a letter for the unknown. Solve.

Answer: $p = 9$ pages; $p \times 6 = 54$ or $54 \div 6 = p$; $(30 \div 6) + (24 \div 6) = 5 + 4 = 9$

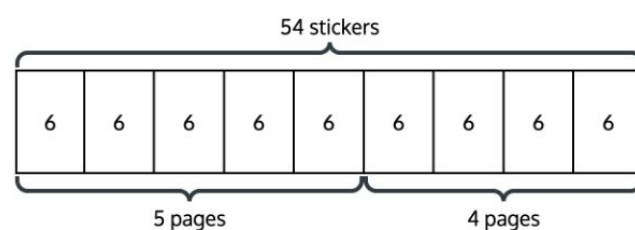
TEACHER GUIDANCE

Ask students what is different about this problem compared to the first one. (We know the total but not the number of groups.)

Guide students through drawing a tape diagram with the total of 54 stickers labeled on top, equal parts each labeled "6 stickers," and p representing the unknown number of pages.

Ask students to write an equation. ($p \times 6 = 54$ or $54 \div 6 = p$.)

Have students solve by decomposing the dividend: $54 \div 6$ as $(30 \div 6) + (24 \div 6) = 5 + 4 = 9$. Ask students what p equals and have them write a sentence to answer the problem.



- 3 **Solve a word problem involving units of 7 by writing an equation with a letter for the unknown, drawing a model, and choosing a strategy to find the answer.**

A parking lot has 6 rows of cars. Each row has 7 cars. How many cars are in the parking lot?

Write an equation using a letter for the unknown. Draw a tape diagram and solve.

Answer: $t = 42$ cars; $6 \times 7 = t$; $(5 \times 7) + (1 \times 7) = 35 + 7 = 42$

Students work independently.

As students are ready, ask them to explain what their letter represents and how they broke apart the problem to solve.

Monitor For

- Does the student correctly identify what the unknown represents — the total, the number of groups, or the size of each group?
- Can the student draw and label a tape diagram that matches the structure of the word problem?
- Does the student decompose factors or dividends into smaller, known facts to solve?

Instructional Moves

- If a student struggles to identify the unknown, ask: "What do you already know? What is the problem asking you to find?" and connect each quantity to the tape diagram.
- When decomposing, prompt students to use fives facts as a starting point: "What is 5×6 ? How many more groups do you need?"
- If a student's tape diagram doesn't match the problem structure, have them reread the problem and point to which part of the diagram shows each quantity.

Reflection Questions

1. In Problem 1, the unknown was the total. In Problem 2, the unknown was the number of groups. How did knowing which quantity was unknown help you decide how to solve?
Answer: When the total was unknown, I multiplied the number of groups by the size of each group. When the number of groups was unknown, I divided the total by the size of each group.
2. How did you break apart the multiplication or division into smaller facts to solve?
Answer: I used fives facts. For example, for 7×6 , I did $5 \times 6 = 30$ and $2 \times 6 = 12$, then added $30 + 12 = 42$.
3. How does the tape diagram help you see the structure of a word problem?
Answer: The tape diagram shows the equal groups, the size of each group, and the total, so I can see which part is unknown.

Mini Lesson Student Materials

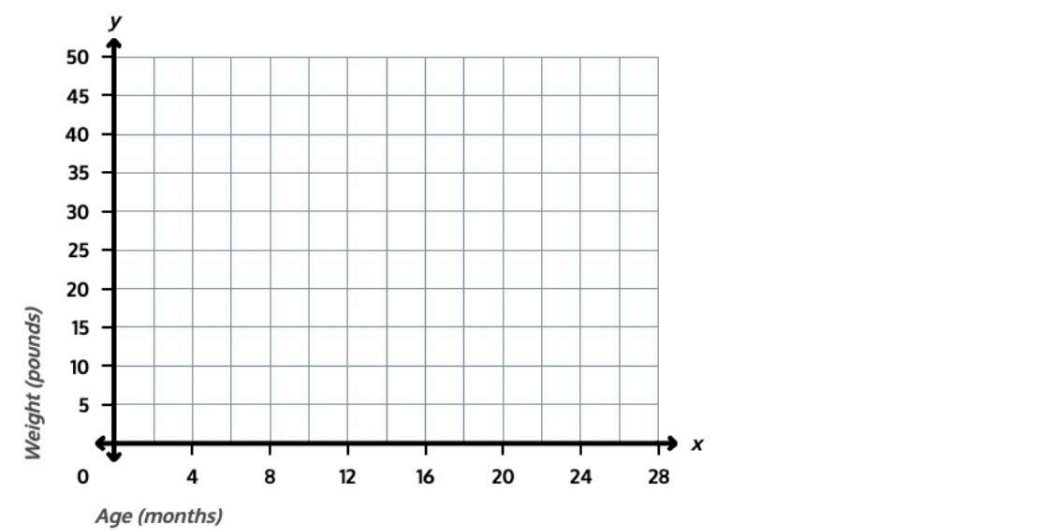
Name _____

1. The table below shows Biscuit the dog's weight at different ages.

Age (months)	Weight (pounds)
0	3
4	15
8	25
12	35
16	40
20	45
24	45
28	45

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Plot each data point on the coordinate plane below. Then, connect the points with line segments to make a line graph.



2. Use the line graph of Biscuit's weight to answer the following questions.

- a. How much weight did Biscuit gain in the first 4 months of his life, between 0 months and 4 months?
- b. How much weight did Biscuit gain in the second 4 months of his life, between 4 months and 8 months?
- c. How much more weight did Biscuit gain in the first 4 months than in the second 4 months?

3. Use the line graph of Biscuit's weight to answer the following questions.

- a. During which 4-month period did Biscuit gain weight the fastest? How can you tell from the graph?
- b. What happened to Biscuit's weight between the ages of 20 months and 28 months? How does the line graph show this?

Mini Lesson Sample Script

Materials

(S) Personal white board

PROBLEM

1

Model a multiplication word problem involving units of 6 by drawing and labeling a tape diagram, then solve by breaking the factor into two smaller facts.

Rosa has 7 boxes of crayons. Each box has 6 crayons. How many crayons does Rosa have in all?

Draw and label a tape diagram to model the problem. Write an equation using a letter for the unknown. Solve.

Answer: $c = 42$ crayons; $7 \times 6 = c$; $(5 \times 6) + (2 \times 6) = 30 + 12 = 42$

SAMPLE SCRIPT

T: Read this problem. What do we know?

S: There are 7 boxes and each box has 6 crayons.

T: What are we trying to find?

S: How many crayons there are in all.

T: We can draw a tape diagram to model this. How many equal parts should we draw?

S: 7, because there are 7 boxes.

T: Right. Draw 7 equal parts on your board. What goes inside each part?

S: 6, because each box has 6 crayons.

T: Label each part "6." Now, the total is what we're trying to find. Let's use the letter c for the total number of crayons. Write " c crayons" above the whole tape.

S: (Label the tape diagram.)

T: Write an equation for this problem.

S: $7 \times 6 = c$.

T: Do you know 7×6 right away? Here's a way to break it into facts you might already know. What if we split the 7 groups into 5 groups and 2 groups? (Draw a line on the tape diagram between the 5th and 6th parts.) What is 5×6 ?

S: 30.

T: And 2×6 ?

S: 12.

T: So what is $30 + 12$?

S: 42.

T: So $c = 42$ crayons. Write a sentence to answer the problem.

S: Rosa has 42 crayons in all.

PROBLEM

2

Model a division word problem involving units of 6 by drawing a tape diagram with an unknown number of groups, then solve using a related multiplication strategy or by decomposing the dividend.

Marcus has 54 stickers. He puts 6 stickers on each page. How many pages does Marcus use?

Draw and label a tape diagram to model the problem. Write an equation using a letter for the unknown. Solve.

*Answer: $p = 9$ pages; $p \times 6 = 54$
or $54 \div 6 = p$; $(30 \div 6) + (24 \div 6)$
 $= 5 + 4 = 9$*

SAMPLE SCRIPT

S: There are 54 stickers total, and each page gets 6 stickers.

T: What are we trying to find?

S: How many pages Marcus uses.

T: So this time, do we know the total or are we looking for it?

S: We know the total. We don't know how many groups.

T: Good. Let's use the letter p for the number of pages. On your tape diagram, label the top "54 stickers." Each part gets "6 stickers." We don't know yet how many parts there are — that's p . Write an equation.

S: $54 \div 6 = p$. $\rightarrow p \times 6 = 54$.

T: Now, can we break 54 into parts that are easier to divide by 6? What's a number you know that divides evenly by 6?

S: 30. $30 \div 6 = 5$.

T: If we've used 30 of the 54 stickers, how many are left?

S: 24.

T: What's $24 \div 6$?

S: 4.

T: So how many pages is that altogether?

S: $5 + 4 = 9$ pages.

T: $p = 9$ pages. Write a sentence to answer the problem.

S: Marcus uses 9 pages.

PROBLEM

3

Solve a word problem involving units of 7 by writing an equation with a letter for the unknown, drawing a model, and choosing a strategy to find the answer.

A parking lot has 6 rows of cars. Each row has 7 cars. How many cars are in the parking lot?

Write an equation using a letter for the unknown. Draw a tape diagram and solve.

Answer: $t = 42$ cars; $6 \times 7 = t$; $(5 \times 7) + (1 \times 7) = 35 + 7 = 42$

SAMPLE SCRIPT

T: Now try this one on your own. Read the problem, write an equation with a letter for the unknown, draw a tape diagram, and solve.

S: (Work.)